DAFI LABS PRIVATE LIMITED

DevOps Engineer Internship Program

Week 1 Assignment Report

Git, GitHub, Vercel, DNS, Environment Variables & AI-Assisted DevOps

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GitHub: github.com/omar2004khaled

Repository: github.com/omar2004khaled/Portfolio

Live URL: portfolio-chi-orcin-82.vercel.app

Date: July 17, 2026

Contents

Assignment Overview

This report documents the completion of the Week 1 DevOps Engineer Internship assignment at Dafi Labs. The objective of the assignment was to build practical, hands-on confidence with the core tools of a modern web deployment workflow: Git and GitHub for version control, Vercel for continuous deployment, DNS for domain resolution, environment variables for configuration security, and AI-assisted tools for accelerated, self-directed learning.

The deliverable is a responsive, single-page developer portfolio built with plain HTML, CSS, and JavaScript (no build tooling or external frameworks), deployed live on Vercel and version-controlled on GitHub across three branches.

Submission Checklist

- ✓ GitHub Repository Link
- ✓ Live Vercel URL
- ✓ Research Document (this report)
- ☐ Deployment Screenshots *(to be inserted — see Task 4 and Task 3 notes)*
- ✓ AI Conversation / Summary (Task 7)

1 Portfolio Website

Task 1 — Create Portfolio *Marks: 30*

A responsive, one-page developer portfolio was built using HTML5, CSS3, and vanilla JavaScript, with AI-assisted development (Cursor AI / ChatGPT) used to accelerate implementation while every feature was reviewed and understood before inclusion. The site runs entirely client-side, with no build step or package installation required.

Key Features

- Fully responsive layout adapting to desktop, tablet, and mobile viewports
- Light / dark mode toggle with persisted user preference via `localStorage`
- Typewriter-style animated hero section cycling through role titles
- Sticky, scroll-aware navigation header with mobile hamburger menu
- Scroll-reveal animations powered by the Intersection Observer API
- Active-section highlighting in the navigation bar as the user scrolls
- Dedicated *Projects* section showcasing four key works
- *Contact* section with direct links (email, phone, location, GitHub, LinkedIn)
- A "View CV" action that opens the resume PDF in a new browser tab
- Back-to-top button for long-page navigation convenience

Technology Stack

Layer	Technology
Markup	HTML5 (semantic sectioning)
Styling	CSS3 (custom properties, Flexbox/Grid)
Interactivity	Vanilla JavaScript (ES6+)
Icons	Font Awesome 6.4.0 (CDN)
Typography	Google Fonts — Outfit & JetBrains Mono

Local Testing

The project was verified by opening `index.html` directly in a browser, as well as through the VS Code "Live Server" extension, confirming that all interactive features (theme toggle, typewriter effect, scroll reveals, mobile navigation) function correctly without a build pipeline.

2 GitHub Repository and Branches

Task 2 — GitHub Repository *Marks: 15*

The project was version-controlled with Git and pushed to a public GitHub repository.

Repository: https://github.com/omar2004khaled/Portfolio

Branching Strategy

Three branches were created to reflect a standard environment-promotion workflow:

Branch	Purpose
`main`	Stable, production-ready codebase; source of truth for deployments
`development`	Active development branch for new features and fixes prior to review
`production/staging`	Pre-production verification branch, mirroring what will be promoted to `main`

Listing 1: Branch creation commands

```
git checkout -b development
git checkout -b production/staging
git push -u origin development
git push -u origin production/staging
```

3 Vercel Deployment

Task 3 — Deploy on Vercel *Marks: 20*

The repository was connected to Vercel, which automatically detected the static site and deployed it without any custom build configuration.

Live URL: https://portfolio-chi-orcin-82.vercel.app/
Status: HTTP 200 — verified reachable and rendering correctly

Deployment Notes

- No build command was required (static HTML/CSS/JS).
- No install command was required (no external dependencies to resolve).
- Vercel's GitHub integration triggers an automatic redeploy on every push to the connected branch, providing continuous deployment out of the box.

Screenshot placeholder: Vercel deployment dashboard showing successful build and live status.

4 DNS Research

Task 4 — DNS Research *Marks: 20*

A custom domain was not purchased for this assignment; the project is served through the free Vercel-provided subdomain. The DNS configuration required to attach a custom domain in the future was researched and is documented below for completeness.

Required DNS Records for a Custom Domain on Vercel

Type	Host	Value	Purpose
A	`@`	`76.76.21.21`	Points the root/apex domain to Vercel's edge network
CNAME	`www`	`cname.vercel-dns.com`	Points the `www` subdomain to Vercel
TXT	(as provided)	Vercel-issued verification string	Proves domain ownership during setup

Screenshots to Include Upon Domain Purchase

1. Vercel project domain settings page
2. DNS record configuration page from the domain registrar
3. Verified/active domain status confirmation in Vercel

Screenshot placeholders: to be inserted here once a custom domain is connected.

5 Environment Variables

Task 5 — Environment Variables *Marks: –*

A sample environment file, `.env.example`, was created to document the expected configuration keys without exposing any real secrets.

Listing 2: .env.example (illustrative structure)

```
# Example environment variables
API_BASE_URL=https://api.example.com
CONTACT_FORM_KEY=your_key_here
```

Why `.env` Should Never Be Committed

- It may contain secret keys, tokens, and private credentials.
- Committing it can expose sensitive information publicly, especially in public repositories.
- It creates security risks if the repository is ever shared, forked, or made public later.

The real `.env` file is excluded from version control via `.gitignore`.

Listing 3: Relevant .gitignore entry

```
.env
```

6 DevOps Research

Task 6 — DevOps Research *Marks: 20*

GitHub

GitHub is a cloud-based platform for hosting Git repositories. It enables version control, collaborative development through pull requests and code review, issue tracking, and branch management — forming the backbone of modern software collaboration workflows.

Vercel

Vercel is a cloud platform purpose-built for deploying frontend and static web applications. It integrates directly with GitHub, automatically building and publishing a project on every push, and provides global CDN distribution, preview deployments, and free HTTPS out of the box.

DNS (Domain Name System)

DNS is the internet's naming system: it translates human-readable domain names into the IP addresses that computers use to locate servers, allowing browsers to find and connect to websites by name rather than numeric address.

Domains

A domain is the human-readable address a user types into a browser (e.g., `example.com`) to reach a website, rather than its underlying IP address.

.gitignore

The `.gitignore` file tells Git which files and directories to exclude from version control. It is typically used to omit local configuration, build artifacts, dependency folders (e.g., `node_modules`), and environment files containing secrets.

Build Command

The build command compiles and prepares a project for deployment (e.g., transpiling, bundling, minifying). This portfolio required no build command, as it is served directly as static HTML, CSS, and JavaScript.

Install Command

The install command fetches and installs a project's dependencies (typically via a package manager such as npm). No install command was needed here, since the project has no external package dependencies.

Environment Variables

Environment variables store configuration values — such as API keys, secrets, and environment-specific settings — outside of the source code, keeping sensitive data out of version control and allowing the same codebase to behave differently across development, staging, and production environments.

7 AI Learning Summary

Task 7 — AI Learning *Marks: –*

Using ChatGPT and Cursor AI as learning aids, the deployment of an application on a Ubuntu server using **Nginx** (as a reverse proxy) and **PM2** (as a Node.js process manager) was researched and understood step by step.

Deployment Steps

1. Update the Ubuntu server's package index and installed packages.
2. Install Node.js and npm.
3. Install Nginx as the web server / reverse proxy.
4. Install PM2 globally to manage the Node.js process.
5. Upload the application code to the server.
6. Start the application using PM2 to keep it running persistently.
7. Configure Nginx to reverse-proxy incoming traffic to the PM2-managed app.
8. Verify the deployment by accessing the server's public address in a browser.

Listing 4: Representative command sequence

```
sudo apt update && sudo apt upgrade -y
curl -fsSL https://deb.nodesource.com/setup_lts.x | sudo -E bash -
sudo apt install -y nodejs nginx
sudo npm install -g pm2

# Deploy app
pm2 start app.js --name my-app
pm2 save
pm2 startup

# Configure Nginx reverse proxy, then:
sudo nginx -t
sudo systemctl reload nginx
```

Reflection

This exercise clarified how source control (GitHub), managed deployment platforms (Vercel), and traditional server administration (Ubuntu, Nginx, PM2) represent complementary — rather than competing — approaches to shipping software, and how AI tools can accelerate learning a new stack while still requiring the developer to understand and verify each command before running it.

Conclusion

All seven tasks of the Week 1 assignment were completed: a responsive portfolio was built and tested locally, version-controlled across three branches on GitHub, and deployed live on Vercel with a verified HTTP 200 response. DNS requirements for a future custom domain were researched and documented, environment variable hygiene was demonstrated through a sample `.env.example` and `.gitignore` entry, core DevOps concepts were explained in the researcher's own words, and a full Ubuntu/Nginx/PM2 deployment workflow was learned with AI assistance and documented above.

Criteria	**Marks**
Project	30
GitHub	15
Deployment	20
Research	20
Originality	15
Total	**100**

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Date: July 17, 2026